Decent Coaching Classes

VIII SB

Mind Your Math - IX

ANSWER KEY & SOLUTIONS

Q.1 (A) MCQ (1 mark each) **[8]**

(1) C (2) B (3) C (4) A (5) A (6) A (7) A (8) B

Hints: (2) The centroid divides a median 2 : 1, so GE is one-third of BE: BE = 3 × 3 = 9 cm. (4) −18 = −9 × 2 and −9 + 2 = −7.

(5) $8x^3 + 27 = (2x)^3 + 3^3 = (2x + 3)(4x^2 - 6x + 9)$. (6) $(x - 2)(x + 2) / [(x - 2)(x + 1)] = (x + 2)/(x + 1)$. (7) x = ky, 20 = 5k, k = 4.

Q.1 (B) Fill in the blanks / True or False (1 mark each) **[8]**

(1) median (2) altitude (3) 5 (4) True (5) $a^2 - ab + b^2$ (6) False ($x^2 - x - 12 = (x - 4)(x + 3)$) (7) constant (8) False (inverse variation)

(3) l(GP) = (1/3) × 15 = 5 cm.

Q.2 Any FIVE (2 marks each) **[10]**

(1) The centroid divides the median in the ratio 2 : 1, so l(PG) : l(GM) = 2 : 1. l(GM) = 8 ÷ 2 = **4 cm**; l(PM) = 8 + 4 = **12 cm**

(2) l(BG) = 2 × l(GE) = 2 × 2.5 = 5 cm. l(BE) = 5 + 2.5 = **7.5 cm**

(3) Factors of 21 with sum −10 are −7 and −3. $x^2 - 7x - 3x + 21 = x(x - 7) - 3(x - 7) =$ **$(x - 7)(x - 3)$**

(4) 3 × 8 = 24; factors with sum 14 are 12 and 2. $3y^2 + 12y + 2y + 8 = 3y(y + 4) + 2(y + 4) =$ **$(y + 4)(3y + 2)$**

(5) $27p^3 + 64 = (3p)^3 + 4^3 =$ **$(3p + 4)(9p^2 - 12p + 16)$**

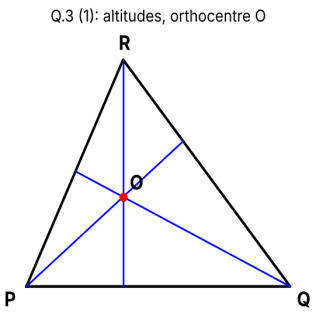
(6) a = kb; 18 = 6k, so k = 3. When b = 15: a = 3 × 15 = **45**

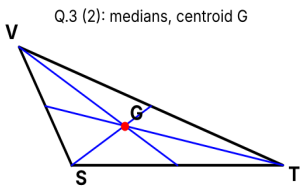
(7) x × y = k = 24 × 5 = 120. When y = 8: x × 8 = 120, so x = **15**

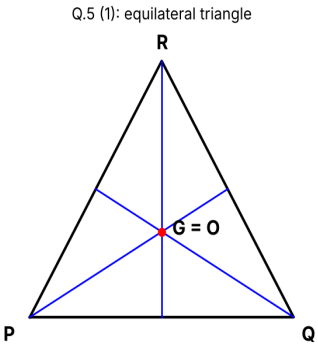
Q.3 Any FOUR (3 marks each) **[12]**

(1) Draw any acute-angled ΔPQR. Using a set-square, draw perpendiculars from P, Q and R to the opposite sides (3 altitudes). They meet at O inside the triangle (orthocentre). [Neat triangle 1, three altitudes 1, O marked 1]

(2) Draw any obtuse-angled ΔSTV. Mark the midpoint of each side and join it to the opposite vertex (3 medians). They meet at G inside the triangle (centroid). [Triangle 1, three medians 1, G marked 1]







(3) 2 × (−12) = −24; factors with sum −5 are −8 and 3. $2x^2 - 8x + 3x - 12 = 2x(x - 4) + 3(x - 4) =$ **$(x - 4)(2x + 3)$**

(4) $250p^3 - 128 = 2(125p^3 - 64) = 2[(5p)^3 - 4^3] =$ **$2(5p - 4)(25p^2 + 20p + 16)$**

(5) $y = kx^2$; 50 = k × 25, so k = 2. When x = 8: y = 2 × 64 = **128**

(6) Pipes and time are in inverse variation: pipes × time = k = 4 × 9 = 36. For 6 pipes: 6 × t = 36, so t = **6 hours**

Q.4 Any THREE (4 marks each) **[12]**

(1) G divides each median in the ratio 2 : 1.
(i) l(AG) = 2 × l(GD), so l(GD) = 10 ÷ 2 = **5 cm** (ii) l(BG) = 2 × l(GE) = 2 × 4 = **8 cm** (iii) l(BE) = 8 + 4 = **12 cm** (iv) l(CG) = (2/3) × 21 = **14 cm** [1 mark each]

(2) $m^2 - 5m + 6 = (m - 2)(m - 3)$; $m^2 - m - 2 = (m - 2)(m + 1)$; $m^2 + m = m(m + 1)$; $m^2 - 9 = (m - 3)(m + 3)$.
Expression = $[(m - 2)(m - 3)] / [(m - 2)(m + 1)] \times [m(m + 1)] / [(m - 3)(m + 3)]$ = **m/(m + 3)**

(3) Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with a = 3p + 2q, b = 3p − 2q: a − b = 4q; $a^2 + ab + b^2 = (9p^2 + 12pq + 4q^2) + (9p^2 - 4q^2) + (9p^2 - 12pq + 4q^2) = 27p^2 + 4q^2$.
Result = $4q(27p^2 + 4q^2)$ = **$108p^2q + 16q^3$** (the same result by direct expansion earns full marks)

(4) $p \propto 1/q^2$, so $p \times q^2 = k$. k = 8 × 9 = 72. (i) q = 6: p × 36 = 72, **p = 2**. (ii) p = 18: 18 × q^2 = 72, q^2 = 4, **q = 2** (or −2).

(5) Speed × time = k = 40 × 9 = 360 (inverse variation). (i) s = 60: 60 × t = 360, **t = 6 hours**. (ii) t = 7.5: s × 7.5 = 360, **s = 48 km/hr**.

Q.5 Any TWO (5 marks each) **[10]**

(1) Draw seg QR = 5 cm; with compass arcs of radius 5 cm from Q and R, locate P; join PQ, PR. Draw the three medians (join midpoints to opposite vertices) and the three altitudes (perpendiculars with set-square). [Triangle 2, medians and altitudes 2]
Observation: in an equilateral triangle every median is also an altitude, so the centroid G and the orthocentre O coincide (they are the same point, inside the triangle). [1]

(2) (i) 12 × (−35) = −420; factors with sum −1 are −21 and 20. $12x^2 - 21x + 20x - 35 = 3x(4x - 7) + 5(4x - 7)$ = **(3x + 5)(4x − 7)** [3]
(ii) $16a^3 - 250 = 2(8a^3 - 125) = 2[(2a)^3 - 5^3]$ = **$2(2a - 5)(4a^2 + 10a + 25)$** [2]

(3) Workers and days are in inverse variation: w × d = k.
(i) k = 18 × 20 = **360** [1] (ii) w × 12 = 360, so **w = 30 workers** [2] (iii) 24 × d = 360, so **d = 15 days** [2]

Note: Award full marks for any correct alternative method.